Date: 28-July-2026 & 29-July-2026

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

COMPUTER NETWORKS (BACSE203)

CLASS ID: VL2026270103931, 3933 **FALL SEM 2026-27**

Exp-3:

Task-1: Construction and Transmission of a Custom IEEE 802.3 Ethernet Frame Carrying a Text Message.

Description:

Develop a program that constructs a custom IEEE 802.3 Ethernet frame manually and transmits it over the local network. The payload of the frame should contain a text message.

Ethernet Frame Format

Field	Size (Bytes)
Destination MAC	6
Source MAC	6
Length	2
LLC Header	3
Payload (Text)	46–1500
Padding (if required)	Variable
FCS	Generated by NIC

The developed program should accept Destination MAC address, Source MAC address and a text message and constructs the ethernet frame manually. Further, ensure the payload is at least 46 bytes (add padding if necessary). The constructed should be transmitted using raw socket and finally capture the custom frame using Wireshark network sniffing tool.

Sample payload

Hello VIT Students

Expected Wireshark output

- Students should verify Destination MAC, Source MAC, Length field, LLC Header, Payload, padding.

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Task-2: Modify the developed program from Task-1 such that it should send multiple text messages and broadcast to all messages.

Expected Wireshark output

- Students should verify the modifications in the Wireshark dissection.

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Task-3:

Construction and Transmission of an Ethernet II Frame Carrying an ARP Request.

Requirements: The developed program should construct the Ethernet II header manually, set ether type as 0x0806 (ARP) and construct the complete ARP request. Finally send the frame using AF_PACKET raw socket. Verify the frame using Wireshark.

Protocol Stack

Ethernet II

- ├── Destination MAC : ff:ff:ff:ff:ff:ff
- ├── Source MAC
- ├── EtherType : 0x0806
- └── ARP
  - ├── Hardware Type
  - ├── Protocol Type
  - ├── Opcode
  - ├── Sender MAC
  - ├── Sender IP
  - ├── Target MAC
  - └── Target IP

ARP Packet Fields

Field	Size
Hardware Type	2 Bytes
Protocol Type	2 Bytes
Hardware Address Length	1 Byte
Protocol Address Length	1 Byte
Opcode	2 Bytes
Sender MAC	6 Bytes
Sender IP	4 Bytes

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Field	Size
Target MAC	6 Bytes
Target IP	4 Bytes